

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0255

Mazomanie Industrial 138 kV
August 16, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0255. The Interconnection Customer proposes a 318.8 MW natural gas combined cycle facility interconnecting at the Mazomanie Industrial 138 kV substation in the Pennsylvania area. The request was received on July 22, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 140 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$5,351,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

WHEELABRATOR FRACKVILLE ENERGY 345/1kV XFMR: 104.7% loading (824 MVA) following the outage of the TALENENERGY MARTINS CRK LLC HARWOOD to KLINE TOWNSHIP COGEN FACILITY 345 KV Line; DFAX = 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WHEELABRATOR FRACKVILLE ENERGY 345/1kV XFMR	Replace transformer with higher-rated unit	\$16,001,000	33% (\$5,351,000)
	Total allocated to project		\$5,351,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.